

MANISTIQUE, MI, USA

WMO: 726399

Lat: 45.950N

Lon: 86.230W

Elev: 584

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 86-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.2	3.2	-11.9	3.0	2.7	-8.0	3.7	4.4	32.4	24.1	28.2	24.3	9.0	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	77.4	64.9	74.8	64.9	72.7	64.0	69.9	74.3	68.3	72.5	66.6	70.6	8.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	105.8	73.3	66.2	98.8	71.3	64.5	92.7	69.1	34.3	74.5	32.9	72.6	31.5	70.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.3	20.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	22.1	20.8	28.4	37.3	48.2	58.0	64.1	64.9	58.8	45.9	36.2	28.5
	DBStd	16.91	9.27	8.62	8.25	5.59	5.56	5.58	4.61	5.04	6.96	6.47	6.66	9.28
	HDD50	4091	864	816	668	380	101	6	0	13	162	414	667	
	HDD65	8201	1329	1236	1133	830	523	220	74	64	203	594	863	1132
	CDD50	1496	0	0	0	0	44	245	436	460	277	33	1	0
	CDD65	131	0	0	0	0	0	9	45	60	17	0	0	0
	CDH74	307	0	0	0	0	5	29	104	161	8	0	0	0
	CDH80	22	0	0	0	0	0	7	5	10	0	0	0	0
	WSAvg	9.4	11.1	10.2	10.0	9.7	8.4	7.1	7.0	8.3	9.2	10.6	11.2	10.5
	PrecAvg	30.7	1.7	1.2	2.0	2.5	3.1	3.1	3.0	3.0	3.5	3.2	2.5	1.9
(15) Precipitation	PrecMax	39.5	5.7	3.2	4.9	4.8	5.6	6.8	6.7	4.4	8.1	5.8	4.8	3.9
	PrecMin	24.4	0.7	0.2	0.3	0.8	1.6	1.0	0.6	0.9	1.1	1.2	0.9	0.2
	PrecStd	3.8	1.1	0.7	1.3	1.0	1.0	1.4	1.6	0.8	1.4	1.3	1.1	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	39.5	47.4	56.4	71.0	77.3	80.3	81.3	74.7	63.2	52.7	52.3
		MCWB	34.9	34.1	40.8	45.4	54.8	63.7	64.5	68.9	66.6	58.4	49.3	37.2
	2%	DB	38.9	36.7	42.7	51.0	63.5	73.0	77.0	78.3	72.3	60.1	49.9	45.1
		MCWB	32.6	33.2	37.0	43.2	52.8	62.5	64.5	66.9	66.1	54.8	46.8	35.5
	5%	DB	36.0	34.7	39.5	47.5	59.6	69.9	74.5	75.2	70.3	57.5	47.4	41.0
		MCWB	32.7	31.8	35.1	41.5	51.1	60.7	64.5	65.5	64.6	52.9	44.5	36.2
	10%	DB	34.2	32.9	37.3	45.3	56.4	66.6	72.3	73.0	68.4	55.4	45.8	38.8
		MCWB	31.7	30.2	33.6	40.2	49.0	59.1	63.4	64.3	63.3	50.9	42.9	35.2
	0.4%	WB	36.5	35.3	41.4	48.1	58.9	67.0	71.4	72.9	70.1	59.4	50.2	40.3
		MCDB	42.2	38.0	46.3	53.2	66.7	74.2	75.7	78.8	72.5	62.8	52.4	43.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.3	33.5	38.0	45.1	54.9	64.4	69.5	70.4	67.6	56.3	47.4	38.4
		MCDB	37.5	36.1	41.2	48.9	60.7	69.8	73.8	75.5	70.4	59.2	48.9	41.9
	5%	WB	33.1	32.2	36.0	42.9	52.4	62.3	67.7	68.8	66.0	53.8	45.6	36.8
		MCDB	35.8	34.7	38.6	46.6	58.0	67.7	71.8	73.3	68.7	56.6	46.8	40.7
	10%	WB	32.0	30.5	34.4	40.7	50.2	60.2	65.9	67.1	64.3	51.4	43.6	35.1
		MCDB	34.6	32.7	36.8	44.9	55.5	66.0	70.1	71.4	67.4	55.1	45.9	38.9
	5% DB	MDBR	11.9	13.4	12.4	11.8	13.8	13.8	13.9	13.9	12.6	10.8	9.6	10.0
		MCDBR	13.5	13.6	13.0	14.6	18.3	17.2	16.4	15.4	12.2	11.8	10.0	10.1
		MCWBR	10.9	10.9	9.9	10.0	11.4	10.6	10.1	9.9	8.3	9.3	9.4	7.5
		MCWBR	11.9	13.1	11.3	13.3	16.6	14.8	13.1	13.4	10.9	11.0	10.0	9.5
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.2	10.8	9.2	9.7	11.2	10.3	9.5	9.6	8.4	9.6	9.6	8.3
	taub		0.269	0.282	0.311	0.355	0.384	0.398	0.394	0.388	0.369	0.335	0.303	0.272
		taud	2.403	2.343	2.378	2.362	2.347	2.342	2.368	2.385	2.436	2.511	2.515	2.449
(42) Clear-Sky Solar Irradiance	Ebn at Noon		268	286	292	286	280	276	275	273	269	264	254	255
		Edh at Noon	22	29	32	36	38	39	37	35	31	25	20	19
	RadAvg		377	668	1094	1441	1766	1907	1945	1647	1240	737	403	290
(45) All-Sky Solar Radiation	RadStd		35	79	106	144	138	145	89	107	91	83	47	26

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page